SAI DEEPAK AMBATI

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building LLM, RAG, and multi-agent AI applications using Python, LangChain, LangGraph, and FastAPI. Skilled in Generative AI, NLP, Machine Learning, vector databases, and REST API development. Passionate about developing practical, scalable AI solutions and applying modern AI technologies to real-world problems.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL, C++

AI / Machine Learning: Machine Learning, Deep Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph

Backend & APIs: FastAPI, Node.js, Express.js, RESTful APIs, JWT Authentication, Microservices Architecture

Databases: MongoDB, MySQL, Redis, Qdrant

Data Analysis: Pandas, NumPy, Streamlit

Tools & Cloud: Git, GitHub, Docker, AWS, Postman

PROJECTS

AgentSpace — Multi-Agent AI Platform [GitHub] | **June 2026 – Present**

Node.js, Express.js, MongoDB, LangChain, LangGraph, Qdrant, Redis, Docker, AWS

- Designed a multi-agent AI platform on the MERN stack with a microservices architecture, enabling orchestration of 8 concurrent AI agent workflows.
- Built a RAG-based document intelligence pipeline using LangChain, LangGraph, and Qdrant, enabling semantic search across 100+ documents with context-aware responses.
- Executed secure REST APIs with JWT authentication, file upload pipelines, and Redis caching, reducing average response latency by 40%.
- Containerized and productionized the application using Docker, implementing a modular, maintainable architecture and configuring AWS deployment aligned with production-grade engineering practices.

ResearchMind — Multi-Agent AI Research System [GitHub] | **Jan 2026 – June 2026**

Python, LangChain, LLMs, Streamlit, Web Scraping

- Architected a 4-agent AI research pipeline using LangChain and LangGraph, coordinating Search, Reader, Writer, and Critic agents with sequential workflows, automatic retries, and rate-limit handling.
- Engineered an AI-powered research pipeline integrating Tavily and BeautifulSoup for real-time web discovery and intelligent content extraction, with LCEL-based workflows for structured Markdown report generation, automated quality scoring, and iterative feedback.
- Designed and implemented a real-time Streamlit dashboard featuring live multi-agent status tracking, session-based research history, and organized views for generated reports, critiques, and source materials.

Cognitive RAG **[GitHub] | Jun 2025 – Dec 2025**

FastAPI, LangChain, ChromaDB, Ollama, Python

- Built a Retrieval-Augmented Generation (RAG) application that enables users to upload PDF documents and ask natural-language questions using locally hosted LLMs through Ollama.
- Implemented an end-to-end document processing pipeline using PyPDFLoader, text chunking, Nomic embeddings, and ChromaDB for semantic retrieval and context-aware response generation.
- Developed FastAPI REST APIs for document ingestion and question answering, integrating document retrieval with locally hosted LLM inference through Ollama.

EDUCATION

Parul University — B. Tech, Computer Science & Engineering (AI) **Sept 2023 – May 2027**

Vadodara, Gujarat CGPA: 8.79 / 10.0

CERTIFICATIONS & TRAINING

- Oracle Cloud Infrastructure (OCI) 2025 Certified Data Science Professional — Oracle, 2025
- MySQL Certification — IIT Indore, 2025
- Generative AI with LangChain and Hugging Face — Udemy, 2026
- Full Stack Web Development — Apna College, 2025
- Python Preparatory — Intellipaat, 2026